

Provider Capacity Optimization & Scheduling Intelligence

Project Overview

Healthcare organizations frequently struggle to balance provider supply with patient demand, resulting in delayed access to care, provider burnout, underutilized appointment capacity, and operational inefficiencies. This project analyzes provider scheduling patterns, appointment utilization, referral demand, and no-show behavior to identify opportunities for improved healthcare access and operational performance.

Using a synthetic multi-location outpatient healthcare dataset, this project evaluates provider utilization, scheduling inefficiencies, appointment wait times, and capacity bottlenecks through advanced SQL analysis and business intelligence reporting.

The ultimate objective is to uncover actionable opportunities to improve scheduling efficiency, maximize provider productivity, reduce patient wait times, and optimize access to care.

Business Problem

Healthcare systems face ongoing challenges in balancing patient demand with provider availability. Operational inefficiencies such as excessive appointment delays, underutilized providers, appointment no-shows, and uneven provider workload distribution can negatively impact patient outcomes, revenue, and overall access to care.

Organizations often lack clear visibility into:

- Which providers are overbooked or underutilized
- Which specialties experience the greatest access delays
- How appointment no-shows impact unused provider capacity
- Whether referral demand exceeds available scheduling capacity

- Which operational adjustments could improve efficiency and patient access

Without proactive capacity analysis, healthcare organizations risk worsening patient access, increased provider strain, preventable revenue loss, and inefficient allocation of clinical resources.

Project Objectives

This project aims to:

1. Evaluate Provider Utilization

Measure provider utilization rates across specialties and locations to identify overutilized and underutilized providers.

2. Identify Scheduling Bottlenecks

Analyze specialty-level and provider-level appointment demand to determine where patient access delays and scheduling inefficiencies occur.

3. Measure Patient Access to Care

Evaluate appointment wait times and referral timelines to assess how quickly patients are receiving care.

4. Quantify Lost Capacity

Determine how appointment cancellations and no-shows contribute to wasted provider availability.

5. Generate Operational Recommendations

Develop data-driven recommendations that healthcare organizations can implement to improve provider efficiency, scheduling performance, and patient access.

Key Questions Being Answered

This project seeks to answer several operational and business-critical questions:

1. Which providers have the highest and lowest utilization rates?
2. Which specialties experience the longest patient wait times?
3. How much provider capacity is lost due to no-shows and cancellations?
4. Are certain clinic locations experiencing greater demand than available capacity?
5. Which providers or specialties may require additional staffing or schedule optimization?
6. What operational interventions could improve patient access and reduce inefficiencies?

Tools & Technologies Used

Data Engineering & Database Design

- SQL
- Relational Database Modeling
- Synthetic Data Generation

Data Analysis

- SQL (CTEs, CASE Statements, Window Functions, Joins, Aggregations)
- Healthcare Operations Analytics

Data Preparation

- Python
- Pandas
- NumPy

Data Visualization

- Power BI (*optional secondary analysis layer*)

Version Control & Documentation

- GitHub
- GitHub Desktop
- Markdown Documentation

Dataset Design

This project uses a synthetic outpatient healthcare operations dataset simulating provider scheduling activity across multiple clinic locations.

Included Data Domains

Providers

- Specialty
- FTE allocation
- Weekly provider availability
- Clinic location

Patients

- Age group
- Insurance type
- Language preference
- Clinical risk level

Appointments

- Appointment type
- Appointment status
- Appointment duration
- Scheduled date vs appointment date
- Cancellation and no-show behavior

Provider Availability

- Daily available provider minutes
- Schedule capacity by date

Referrals

- Referral priority
- Required appointment timeline
- Specialty demand patterns

Methodology

This project follows a healthcare operations analytics workflow:

1. Design relational database schema
2. Generate realistic synthetic healthcare operations data
3. Load and structure relational datasets in SQL
4. Perform provider utilization and scheduling analysis
5. Identify operational bottlenecks using SQL
6. Summarize business findings
7. Develop actionable recommendations

Key Performance Metrics (KPIs)

The following metrics will be analyzed:

- Provider Utilization Rate
- No-Show Rate
- Cancellation Rate
- Average Appointment Wait Time
- Referral-to-Appointment Delay
- Unused Provider Capacity
- Demand-to-Capacity Ratio
- Specialty Bottleneck Analysis

Expected Business Impact

The findings from this analysis are intended to support operational decision-making in outpatient healthcare organizations by helping leadership:

- Improve patient access to care
- Reduce appointment delays
- Optimize provider schedules
- Minimize unused provider capacity
- Improve resource allocation decisions
- Reduce operational inefficiencies

Initial SQL Analysis / Preliminary Findings

Provider Utilization Analysis

The first SQL analysis evaluated provider utilization across specialties and clinic locations by comparing completed appointment minutes against total available provider capacity.

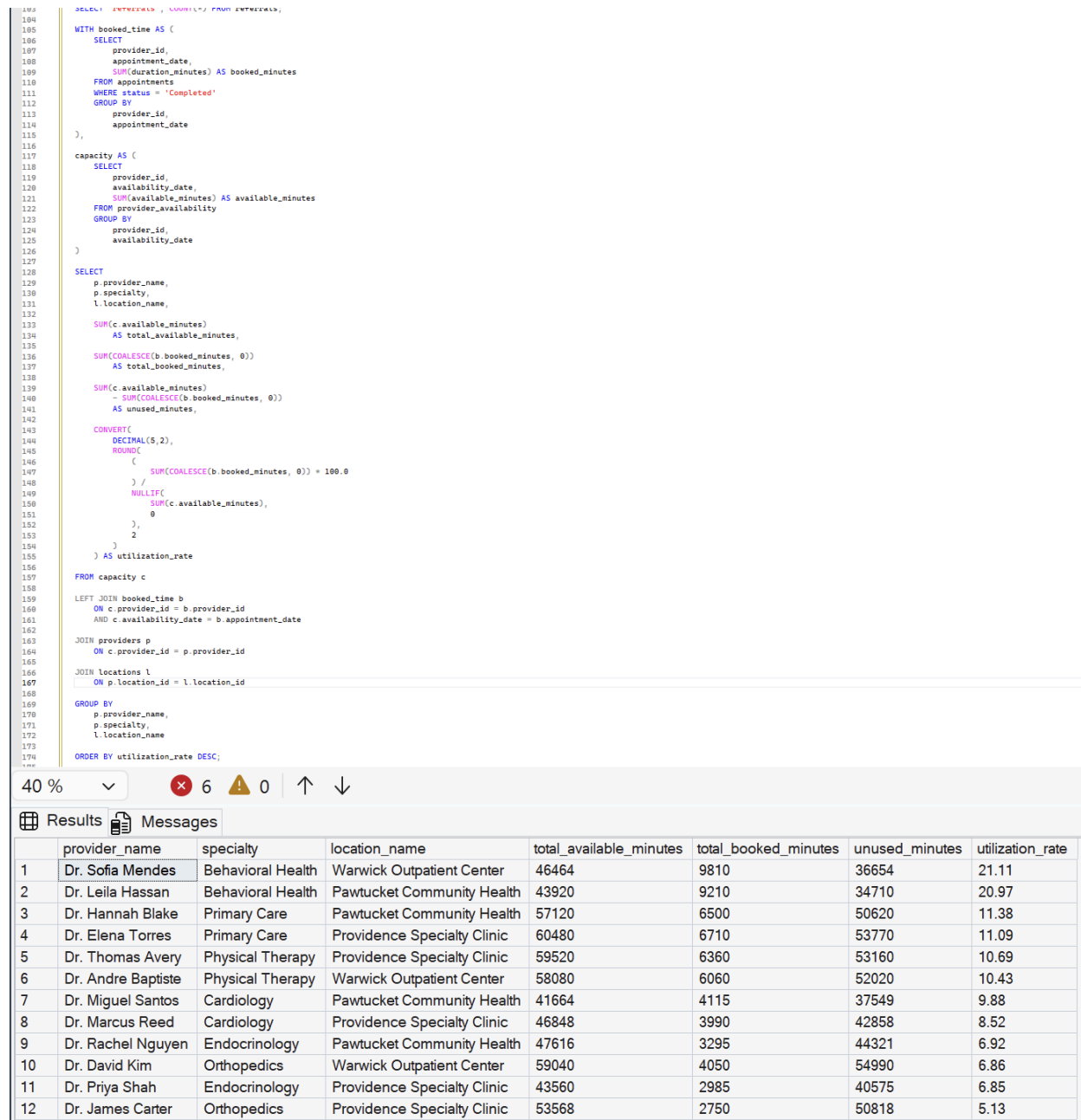
Preliminary findings indicate that **Behavioral Health providers** demonstrated the highest utilization rates across the outpatient network. Dr. Sofia Mendes (21.11%) and Dr. Leila Hassan (20.97%) showed the highest proportion of booked appointment time relative to available provider capacity.

Primary Care and Physical Therapy providers demonstrated moderate utilization, while Orthopedics and Endocrinology showed comparatively lower utilization, suggesting unused scheduling capacity or lower relative patient demand.

These early findings may indicate stronger patient demand for Behavioral Health services relative to available provider supply. Conversely, lower-utilization specialties may present opportunities for scheduling optimization, referral balancing, or provider schedule adjustments to improve operational efficiency.

Further analyses will evaluate appointment wait times, referral delays, no-show behavior, and location-level demand to better understand drivers of utilization and capacity constraints.

Figure 1. Provider Utilization Analysis by Specialty and Location



Wait Time Analysis by Specialty

A second SQL analysis evaluated patient access to care by measuring appointment wait times across specialties using the number of days between appointment scheduling and completed visit dates.

Behavioral Health demonstrated the longest average wait times across the outpatient network, averaging 46 days between scheduling and completed appointments, with some appointments reaching delays of up to 74 days. Endocrinology also showed elevated access delays, averaging 38 days.

Cardiology demonstrated moderate appointment delays (28 days), while Primary Care, Physical Therapy, and Orthopedics maintained comparatively shorter average wait times ranging from 17–18 days.

These findings suggest Behavioral Health and Endocrinology may be experiencing greater demand relative to provider capacity, potentially signaling scheduling bottlenecks, provider shortages, or increased patient demand. Lower wait times among Primary Care and Orthopedics may indicate more balanced provider availability or lower specialty demand.

Further analyses will evaluate whether provider utilization, referral demand, and no-show behavior contribute to these observed access constraints

Figure 2. Wait Time Analysis by Specialty

```

175
176  ✓ SELECT
177      p.specialty,
178
179      COUNT(*) AS total_appointments,
180
181      CONVERT(
182          DECIMAL(5,2),
183          AVG(
184              DATEDIFF(
185                  DAY,
186                  a.scheduled_date,
187                  a.appointment_date
188              )
189          )
190      ) AS avg_wait_days,
191
192      MIN(
193          DATEDIFF(
194              DAY,
195              a.scheduled_date,
196              a.appointment_date
197          )
198      ) AS min_wait_days,
199
200      MAX(
201          DATEDIFF(
202              DAY,
203              a.scheduled_date,
204              a.appointment_date
205          )
206      ) AS max_wait_days
207
208  FROM appointments a
209
210  JOIN providers p
211      ON a.provider_id = p.provider_id
212
213  WHERE a.status = 'Completed'
214
215  GROUP BY p.specialty
216
217  ORDER BY avg_wait_days DESC;

```

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Results



Messages

	specialty	total_appointments	avg_wait_days	min_wait_days	max_wait_days
1	Behavioral Health	406	46.00	21	74
2	Endocrinology	206	38.00	18	59
3	Cardiology	244	28.00	14	44
4	Physical Therapy	357	18.00	3	34
5	Primary Care	567	18.00	3	34
6	Orthopedics	256	17.00	3	34

No-Show Analysis by Insurance Type

A third SQL analysis evaluated appointment no-show behavior across insurance populations to better understand how missed visits may contribute to lost provider capacity.

Medicaid patients demonstrated the highest no-show rate at 15.84%, followed by Medicare patients at 11.72%. Commercially insured patients experienced lower no-show rates (9.49%), while Self-Pay patients demonstrated the lowest observed rate (8.60%).

These findings may suggest that payer-specific access barriers influence appointment attendance. Higher no-show rates among Medicaid populations may reflect transportation limitations, socioeconomic challenges, scheduling instability, or broader healthcare access barriers.

From an operational perspective, elevated no-show rates contribute directly to unused provider capacity and may worsen appointment availability constraints, particularly in already high-demand specialties. Organizations may benefit from targeted interventions such as reminder systems, transportation support, telehealth options, or flexible scheduling strategies for higher-risk patient populations

Figure 3. No-Show Analysis by Insurance Type

218	
219	SELECT
220	pt.insurance_type,
221	
222	COUNT(*) AS total_appointments,
223	
224	SUM(
225	CASE
226	WHEN a.status = 'No-Show'
227	THEN 1
228	ELSE 0
229	END
230	) AS no_show_count,
231	
232	CONVERT(
233	DECIMAL(5,2),
234	(
235	SUM(
236	CASE
237	WHEN a.status = 'No-Show'
238	THEN 1
239	ELSE 0
240	END
241	) * 100.0
242	) / COUNT(*)
243	) AS no_show_rate
244	
245	FROM appointments a
246	
247	JOIN patients pt
248	ON a.patient_id = pt.patient_id
249	
250	GROUP BY pt.insurance_type
251	
252	ORDER BY no_show_rate DESC;

70 %

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Results

Messages

	insurance_type	total_appointments	no_show_count	no_show_rate
1	Medicaid	606	96	15.84
2	Medicare	657	77	11.72
3	Commercial	1244	118	9.49
4	Self-Pay	93	8	8.60

No-Show Analysis by Specialty

A fourth SQL analysis evaluated appointment no-show behavior across clinical specialties to better understand how missed visits may contribute to unused provider capacity and operational inefficiencies.

Behavioral Health demonstrated the highest no-show rate across specialties at 12.89%, followed closely by Orthopedics (12.50%) and Endocrinology (12.30%). Physical Therapy and Primary Care exhibited moderate no-show rates (11.91% and 11.08%, respectively), while Cardiology demonstrated the lowest observed no-show burden at 7.53%.

When considered alongside prior findings, Behavioral Health appears to represent the most operationally constrained specialty within the outpatient network. Behavioral Health providers demonstrated the highest utilization rates, the longest patient wait times (46 days on average), and the highest appointment no-show burden, suggesting a compounded scheduling and access challenge.

Conversely, Orthopedics demonstrated relatively shorter appointment wait times despite elevated no-show rates and lower utilization, potentially indicating underused provider capacity influenced by appointment nonattendance rather than insufficient scheduling availability.

These findings suggest specialty-specific operational strategies may be necessary to improve appointment adherence and maximize provider efficiency. Potential interventions may include automated reminders, telehealth expansion, waitlist optimization, overbooking strategies for high no-show specialties, and targeted scheduling interventions for vulnerable patient populations.

254	SELECT
255	p.specialty,
256	
257	COUNT(*) AS total_appointments,
258	
259	SUM(
260	CASE
261	WHEN a.status = 'No-Show'
262	THEN 1
263	ELSE 0
264	END
265	) AS no_show_count,
266	
267	CONVERT(
268	DECIMAL(5,2),
269	(
270	SUM(
271	CASE
272	WHEN a.status = 'No-Show'
273	THEN 1
274	ELSE 0
275	END
276	) * 100.0
277	) / COUNT(*)
278	) AS no_show_rate
279	
280	FROM appointments a
281	
282	JOIN providers p
283	ON a.provider_id = p.provider_id
284	
285	GROUP BY p.specialty
286	
287	ORDER BY no_show_rate DESC;

80 %	6	0	↑	↓
Results Messages				
	specialty	total_appointments	no_show_count	no_show_rate
1	Behavioral Health	512	66	12.89
2	Orthopedics	352	44	12.50
3	Endocrinology	252	31	12.30
4	Physical Therapy	470	56	11.91
5	Primary Care	722	80	11.08
6	Cardiology	292	22	7.53

Bottleneck Severity & Capacity Risk Analysis

A final SQL analysis combined provider utilization, appointment wait times, no-show behavior, referral demand, and unused provider capacity into a unified specialty-level operational assessment framework.

Behavioral Health emerged as the highest operational risk specialty across the outpatient network, demonstrating the longest appointment delays (46 days), the highest provider utilization, an elevated no-show burden (12.89%), and substantial referral demand. These findings suggest Behavioral Health may be experiencing compounded scheduling strain driven by elevated patient demand relative to available provider capacity.

Endocrinology demonstrated similarly elevated wait times (38 days) and increased no-show rates despite comparatively lower utilization, potentially indicating specialist access limitations or insufficient provider availability rather than scheduling inefficiency alone.

Cardiology demonstrated moderate wait times but comparatively stable appointment adherence, suggesting more balanced operational performance despite moderate demand pressure.

Primary Care, Physical Therapy, and Orthopedics demonstrated shorter wait times and comparatively lower utilization rates, though elevated unused provider minutes suggest opportunities for scheduling optimization and improved provider capacity utilization.

Taken together, these findings highlight the importance of specialty-specific operational strategies, as scheduling inefficiencies and access barriers appear to vary considerably across clinical service lines.

Summary Table

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Results

Messages

	specialty	utilization_rate	avg_wait_days	no_show_rate	total_referrals	avg_referral_window_days	unused_provider_minutes	bottleneck_severity
1	Behavioral Health	11.89	46.00	12.89	151	34.00	140940	Moderate Bottleneck Risk
2	Endocrinology	5.42	38.00	12.30	63	34.00	109496	Moderate Bottleneck Risk
3	Cardiology	6.96	28.00	7.53	92	33.00	108391	Moderate Bottleneck Risk
4	Physical Therapy	6.60	18.00	11.91	132	35.00	175740	Stable / Opportunity Area
5	Primary Care	4.86	18.00	11.08	238	34.00	258470	Stable / Opportunity Area
6	Orthopedics	4.33	17.00	12.50	124	33.00	150256	Stable / Opportunity Area

Recommendations

Based on provider utilization, appointment wait times, referral demand, no-show behavior, and unused provider capacity analyses, several operational recommendations emerge to improve patient access and scheduling performance across the outpatient network.

1. Expand Behavioral Health Scheduling Capacity

Behavioral Health demonstrated the strongest indicators of operational strain, including the highest provider utilization, longest appointment wait times (46 days), elevated no-show rates (12.89%), and substantial referral demand.

Healthcare leadership should consider expanding Behavioral Health scheduling capacity through:

- Additional provider staffing or contracted behavioral health support
- Extended clinic hours or telehealth availability
- Reserved urgent-access appointment slots
- Improved referral prioritization workflows

Reducing Behavioral Health access delays may improve continuity of care, reduce patient disengagement, and improve clinical outcomes.

2. Improve Appointment Adherence for High No-Show Populations

Medicaid patients demonstrated the highest no-show rate (15.84%), while Behavioral Health, Orthopedics, and Endocrinology demonstrated the highest specialty-level nonattendance rates.

Organizations may benefit from targeted interventions, including:

- Automated text and phone reminders
- Same-day confirmation systems
- Transportation assistance partnerships
- Flexible rescheduling workflows
- Expanded telehealth access for vulnerable populations

Reducing appointment nonattendance may improve provider productivity while increasing available appointment access without requiring additional staffing.

3. Optimize Underutilized Provider Capacity

Primary Care, Orthopedics, and Physical Therapy demonstrated comparatively lower provider utilization and substantial unused provider minutes.

Healthcare organizations should evaluate:

- Referral redistribution strategies across specialties
- Schedule optimization for underbooked providers
- Strategic overbooking models in high no-show specialties
- Same-week scheduling opportunities to improve appointment fill rates

Improving provider utilization may reduce operational waste while increasing patient access.

4. Prioritize Endocrinology Access Improvements

Endocrinology demonstrated prolonged appointment delays (38 days) despite comparatively lower utilization rates, suggesting access barriers may stem from provider availability limitations or appointment scheduling inefficiencies.

Potential operational strategies include:

- Additional specialist coverage
- Referral triage optimization
- Chronic disease prioritization workflows
- Capacity balancing across clinic locations

Improved endocrinology access may reduce delays in chronic disease management, particularly for diabetes-related care.

5. Develop Specialty-Level Scheduling Strategies

Findings suggest operational performance varies significantly across specialties, indicating a standardized scheduling approach may not effectively address specialty-specific access barriers.

Healthcare organizations should consider implementing:

- Specialty-specific scheduling rules
- Waitlist optimization programs
- Capacity forecasting using historical demand patterns
- Referral demand monitoring dashboards

A more targeted scheduling model may improve both patient access and provider efficiency across outpatient service lines.

Final Conclusion

This analysis identified meaningful variation in provider utilization, appointment access, no-show behavior, and referral demand across specialties within the outpatient network.

Behavioral Health emerged as the greatest operational bottleneck, while Primary Care, Orthopedics, and Physical Therapy demonstrated opportunities for improved provider capacity utilization.

These findings suggest healthcare organizations may improve operational performance through targeted scheduling interventions, specialty-specific capacity planning, and patient engagement strategies aimed at reducing preventable appointment nonattendance